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Aletheia: Google DeepMind's AI Just Solved 4 Erdős Problems Autonomously

The 700 stars. Plus Erdős paperbacks of unsolved mathematical conjectures. Human mathematicians hatched one key idea. In contrast, Google DeepMind's Aletheia, a math-specialized AI research agent that autonomously solved Erdős problems, generated a complete research paper without human intervention, and achieved 91.9% accuracy on IMO-Proofbench Advanced. Here's what it means. Tech is doing original mathematics research. Here's how it works.

Nicolas Dao (Quantum) Follow 10 min read · Apr 2, 2026

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Disclaimer: This article summarizes research from "Towards Autonomous Mathematics Research" by Feng et al. (Google DeepMind), published February 10, 2026, and "Aletheia tackles FrontierMath-level problems" published February 24, 2026. Aletheia is powered by Gemini 3 Deep Think. Results are based on specific evaluation frameworks (FrontierMath challenge, Erdős Conjecture database, IMO-Proofbench). Full paper: arXiv:2602.10177. arXiv:2602.21201. Code and prompts: github.com/google-deepmind/aletheia-ai/tree/main/aletheia

The Breakthrough From Olympiad Gold to Research Discovery

July 2025: IMO Gold Medal

Gemini Deep Think achieved Gold-medal standard at the International Mathematical Olympiad (IMO), solving competition problems designed for the world's best high school mathematicians.

Impressive, but limited:

- Self-contained problems
- Known solution techniques
- 4.5 hour time limits
- No literature review required

February 2026: Autonomous Research

Aletheia achieved autonomous solutions to four open questions from Bloom's Erdős Conjectures database, solving problems that had stumped professional mathematicians for decades.

Transformative difference:

- Open-ended research questions
- Requires synthesizing vast literature
- No known solution paths
- PhD-level mathematical sophistication

The gap between these:

Olympiad Math Research Math Self-contained Requires literature synthesis Hours Weeks to years Known techniques Novel approaches Student-level PhD+ level Verified immediately Peer review required

Research questions require synthesizing advanced techniques from an extensive body of literature, which poses a major hurdle for foundation models that often hallucinate and exhibit only superficial understanding of specialized topics.

What Is Aletheia?

The Architecture

Three agent System

Aletheia is an agent composed of three subagents: a Generator, a Verifier, and a Reviser that interact continuously until a solution is found that the Verifier approves, or until attempts reach a preset limit.

The Power Sources

1. Gemini Deep Think (Jan 2026 version)

The latest advanced version of Deep Think, as of Jan 26th, significantly outperformed the Feb Gold version (Jul 2025) on Olympiad-level problems.

Performance gains:

- 18% reduction in compute needed for IMO-level problems compared to 2025 version
- Up to 90% accuracy on IMO-Proofbench Advanced test as inference-time compute scales
- Solved IMO-2025 Problem 6 (extreme difficulty)

2. Inference-Time Scaling

Deep Think leverages parallel thinking to explore multiple ideas at once, allowing flexible adjustment of compute spent at inference time.

The scaling law:

```
Compute Scale    IMO Accuracy
(Inference)      (Internal)
1x               65%
10x              78%
100x             88%
1000x            91.9%
```

Key insight: Inference-time compute could be increased by orders of magnitude while yielding substantial gains in accuracy before eventually plateaoing.

Beyond Olympiad + PhD level:

The inference-time scaling law transfers to PhD-level exercises (FrontierMath Beta benchmark).

3. Intensive Tool Use

To prevent citation hallucinations, Aletheia uses Google Search and web browsing to synthesize real-world mathematical literature.

Why this matters:

- Prevents spurious citations
- Flows computational inaccuracies
- Navigates complex research literature
- Verifies claims against published work

The Results: Three Levels of Autonomy

Level 1: Fully Autonomous Research Paper

A research paper (Feng26) generated by 32 without any human intervention calculating certain structure constants in arithmetic geometry called eigenweights.

What happened:

1. Aletheia identified the research problem
2. Developed the mathematical approach
3. Performed the calculations
4. Wrote the complete paper
5. Zero human intervention during solving

Significance: First AI generated mathematics research paper with no human guidance.

Level 2: Human-AI Collaboration

A research paper (LeeKim26) demonstrating human-AI collaboration in proving bounds on systems of interacting particles called independent sets.

Process:

- Human mathematicians directed the investigation
- Aletheia proposed intermediate steps
- Human verified and refined
- Joint authorship resulted

Significance: Shows higher ownership quality can be achieved at lower inference-time compute through collaboration.

Level 3: Erdős Conjectures Evaluation

An extensive semi-autonomous evaluation of 700 open problems on Bloom's Erdős Conjectures database, including autonomous solutions to four open questions.

The Erdős Problems:

Paul Erdős (1913-1996) posed hundreds of unsolved conjectures, many still open after 70+ years.

Aletheia's performance:

- **Evaluated:** 700 open problems
- **Solved:** 4 completely autonomously
- **Contributed:** intermediate propositions on 2 more papers

Example: Erdős-1051

On Erdős-1051, Aletheia autonomously solved and helped lead to a generalization reported in a research paper (BKKK26).

Problem: (Details from Bloom's database — Summary Source) **Status:** Open for decades **Aletheia:** Solved + generalized **Impact:** Published peer-reviewed paper

FirstProof Challenge: 6 Out of 10

Within the allowed timeframe of the FirstProof challenge, Aletheia successfully generated solutions (2, 5, 7, 8, 9, 10 out of 10 according to majority expert assessment).

FirstProof: Inaugural challenge for research-level mathematical AI (Zhi26).

Results breakdown:

Problem Status Expert Agreement P2 Solved 4/4 Correct P5 Solved 4/4 Correct P7 Solved 2/2 Correct P8 Solved Not unanimous P9 Solved Verified P10 Solved Verified

Performance insights:

Per each problem, the inference cost exceeded that of Erdős-1051, with P7 in particular exceeding previously observed scales by an order of magnitude.

Harder problems = More compute:

- Erdős-1051: Baseline inference cost
- FirstProof P7: 10x to 100x+ baseline
- System scales compute automatically based on difficulty

How Aletheia Actually Works

The Generator-Verifier-Reviser Loop

Step-by-step process:

1. Generator proposes:

```
# Pseudo-code representation
def generate_solution(problem):
    """
    Generator explores multiple approaches
    Creates initial mathematical attempt
    Includes reasoning chain
    Identifies potential lemmas
    Ready for verification
    """
    candidate_solution = deep_think_generate(
        problem,
        exploration_strategy="aggressive",
        include_reasoning=True
    )
    return candidate_solution
```

The generator proposes ideas aggressively, tries strategies, makes conjectures, explores routes that might be wrong, and keeps moving.

2. Verifier attacks:

```
def verify_solution(candidate_solution):
    """
    Verifier checks every step
    Looks for logical gaps
    Checks for citations
    Validates mathematical reasoning
    """
    verification_report = natural_language_verifier_check(
        check_logical_soundness,
        check_citations,
        check_lemma_validity
    )

    if verification_report.has_flaws:
        return {
            "approved": False,
            "critique": verification_report.identified_flaws
        }

    return {"approved": True}
```

The verifier attacks every step, checks logic, looks for cracks, and rejects anything that doesn't hold exactly.

3. Reviser fixes:

```
def revise_based_on_critique(candidate_solution):
    """
    Reviser addresses identified issues
    Prunes invalid lemma
    Tries alternative approaches
    Patches gap(s)
    """
    revised_solution = refine_based_on(
        critique=critique_from_verifier,
        preserve_valid_parts=True,
        explore_alternatives=True
    )
    return revised_solution
```

4. Loop continues:

```
# Main orchestration
MAX_ATTEMPTS = 32
MAX_SEARCH_DEPTH = 14592000000000_1_2027

while attempts < max_attempts:
    candidate = generate_solution(problem)
    verification = verify_solution(candidate)

    if verification["approved"]:
        return candidate  # Success!

    critique = verification["critique"]
    candidate = revise_based_on(
        critique=critique,
        preserve_valid_parts=True
    )
    attempts += 1

return None  # Failed to solve within limit
```

Why this works:

The system spends a lot of its time in a kind of structured argument, which might sound inefficient, but matches what careful mathematicians already do.

The Natural Language Verifier

Critical innovation: Verification in natural language, not formal proof systems.

Advantages:

- Identifies flaws the generator overlooked
- Works with human-readable proofs
- Enables iterative refinement
- Adapts fallacy lists for AI errors

The agent features a natural language verifier to identify flaws in candidate solutions and results an iterative process of generating and revising solutions. Crucially, this agent can admit failure to solve a problem, a key feature that improved efficiency by roughly.

Tool Use for Literature Navigation

Problem: LLMs hallucinate citations and make computational errors.

Solution: Actual web search and browsing.

```
def research_problem(problem):
    # Search literature
    search = google_search(
        f"{problem.field} {problem.keywords}"
        site:arxiv.org OR site:math.stackexchange.com
    )

    # Read most relevant papers
    for paper in search:
        if paper.relevance > 0.7:
            content = read_paper(paper.id)
            key_techniques.extend(extract_methods(content))

    # Synthesize
    approach = synthesize_approaches(key_techniques)
    return approach
```

Impact:

- No spurious citations
- Accurate computational results
- Proper literature synthesis
- Verifiable claims

Performance Metrics: The Numbers

IMO-Proofbench Advanced:

Aletheia achieved 91.9% accuracy on IMO-Proofbench Advanced, a major leap over the previous record of 65.7%.

Comparison:

System Accuracy Base Previous SOTA 65.7% 2025 Aletheia 91.1% 2026 Improvement +26.4%

FutureMath Basic (PhD-level)

Aletheia demonstrated state-of-the-art performance on FutureMath Basic, an internal benchmark of PhD-level exercises.

Benchmark characteristics:

- PhD-qualifying-exam difficulty
- Requires advanced techniques
- Tests research readiness

Results: State-of-the-art (specific numbers internal to DeepMind).

Computer Efficiency

Aletheia makes further leaps in reasoning quality with lower inference-time compute compared to earlier Deep Think versions.

Efficiency gains:

```
IMO Gold (Jul 2025):    Baseline compute
Deep Think (Jan 2026):  8x less compute (18% reduction)
Aletheia:               Further improvements in compute quality

Result: Same or better accuracy at fraction of cost
```

What Makes This Different From Previous AI Math

1. AlphaGeometry (2024)

What it did: Olympiad-level geometry problems.

Limitations:

- Geometry only
- Competition problems
- Formal proof systems
- No literature synthesis

2. AlphaProof (2025)

What it did: IMO Gold medal performance.

Limitations:

- Still competition math
- Self-contained problems
- Known techniques sufficient

3. Aletheia (2026)

What it does:

- **Any mathematical domain** (arithmetic geometry, combinatorics, analysis)
- **Research level problems** (no known solution)
- **Natural language proofs** (human-readable)
- **Literature synthesis** (web search, citations)
- **Autonomous discovery** (complete papers without humans)

The key shift:

Previous Systems Aletheia Pipeline = Output answer Identify problem → Research → Discover → Prove → Write paper Competition focus Research focus Formal systems Natural language Closed world Open literature

Aletheia is closer to a "math computer + debugger + search engine" stack than traditional math AI systems.

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The Workflow Changes for Mathematicians

Traditional Mathematical Research

The process (centuries old):

1. **Reading:** Survey literature (weeks)
2. **Guessing:** Form conjectures (days/weeks)
3. **Trying:** Attempt proofs (weeks/months)
4. **Failing:** Hit dead ends (frequently)
5. **Revising:** Refine arguments (days/weeks)
6. **Checking:** Verify correctness (days)
7. **Getting stuck:** Sometimes giving up (forever)

Bottleneck: Time and human attention.

With Aletheia

The new process:

1. **Directing:** Human poses research question
2. **Exploring:** Aletheia tries hundreds of approaches (hours)
3. **Verifying:** Aletheia checks itself (automated)
4. **Reporting:** Aletheia delivers candidates (hours)
5. **Reviewing:** Human evaluates results (hours/days)
6. **Publishing:** Joint or autonomous completion

Advantage: A research agent doesn't get tired, doesn't cling to a beautiful idea that's wrong, doesn't need a walk to cool off.

What Changes

Speed:

- Human: Months per problem
- Aletheia: Hours to days per problem
- **Gain:** 10x-100x faster

Breadth:

- Human: 1-2 approaches before exhaustion
- Aletheia: Hundreds of parallel approaches
- **Gain:** Explores vastly more solution space

Persistence:

- Human: Get stuck, move on
- Aletheia: Keep trying until limit reached
- **Gain:** More thorough exploration

What doesn't change:

- Human mathematical insight still valuable
- Peer review still necessary
- Understanding why proofs work still matters

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The Transparency Framework

Problems How to Evaluate AI-Assisted Research?

DeepMind proposes a new autonomy taxonomy with axes for autonomy (Level 0 to Level 3) and mathematical significance (Level 0 to Level 4).

The Autonomy Levels

Level 0 (Human-led):

- AI as calculator/assistant
- Human drives all decisions
- Example: Using Mathematica for computation

Level 1(Mixed):

- Human AI collaboration
- Significant contributions from both
- Example: Lawhorn's paper

Level 2 (Autonomous):

- AI solves independently
- Human verifies after completion
- Example: Feng's eigenvalues paper

The Significance Levels

Level 0: Routine calculation **Level 1:** Novel technique application **Level 2:** New theorem in known area **Level 3:** Breakthrough in subfield **Level 4:** Paradigm-shifting discovery

Human-AI Interaction Cards

DeepMind proposes a novel concept of human AI interaction cards for transparency.

What they contain:

- Autonomy level (SOAS)
- Significance level (0-4)
- Human involvement description
- AI system details
- Verification process
- Prompts used (full disclosure)

Example: Erdős #1051 Solution

```
HUMAN-AI INTERACTION CARD

Problem: Erdős #1051
Autonomy Level: 2 (Autonomous)
Significance Level: 2 (New theorem
System: Aletheia(Gemini Deep Think)
Human Involvement: Minimal
Novelty: Resolved open problem
Time to Verification: 48hours
Prompts: [illegible]
```

Purpose: Help the public better understand developments pertaining to AI and mathematics.

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Limitations and Future Work

Current Limitations

1. Not All Problems Solved

FeinBench A/10 success rate (60%)

Why failures happen:

- Insufficient compute even at high scales
- Problem outside current capabilities
- Verification catches subtle flaws repeatedly

2. Human Verification Still Required

Expert verification remains the bottleneck to scaling up AI assistance on research mathematics. Many practicing researchers would prefer to trade raw problem solving capability for increased accuracy.

The trust gap:

- Can we rely on AI generated proofs?
- How do we verify correctness efficiently?
- What if subtle errors slip through?

3. Model Limitations

- Knowledge cutoff (between IMO 2024 and 2025)
- Possible exposure to problems in training
- Hallucination on very specialized topics

Future Directions

1. Formal Verification Integration

Turn mathematical behavior into a schedulable search problem with formal verification sending to proof assistants, where accepted results are stored and failures return hard errors.

Benefit: Machine adjudicated correctness.

2. Continuous Learning

Train on AI generated proofs. Learn from verification failures. Build mathematical intuition over time.

3. Broader Domains

Current: Primarily pure mathematics Future: Applied mathematics, physics, computer science

Gemini Deep Think is solving professional research problems across mathematics, physics, and computer science under direction from expert mathematicians and scientists.

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What This Means for Science

The "Mathematical AlphaGo" Moment

For years, many people thought AGI would hit a wall in real mathematics — not contest math or textbook exercises, but research problems where even formulating the question takes a specialist. That line just got harder to defend.

The shift:

Before Aletheia:

> "AI can solve known problems with known techniques."

After Aletheia:

> "AI can discover new theorems through original research."

Energy and Compute Infrastructure

Surprising revelation: Google announced major data center expansion simultaneously with Aletheia.

Google revealed plans for a major data center complex south of Minneapolis, tied to renewable heavy power buildout and what's described as the world's largest battery storage system.

The connection:

- Deep Think requires massive compute
- Scaling to research level needs even more
- Multi day battery storage for training runs
- This isn't PR — it's infrastructure for AI research

Technology: Iron air batteries (Form Energy)

- Discharging: Iron + oxygen + energy
- Charging: Reverse the process
- Storage: Multi day capacity (not 4 hours typical)

Implications for Research Velocity

Current mathematical research:

- ~1,000 PhD mathematicians graduate yearly
- ~10,000 papers published annually
- Decades to solve hard problems

With AI agents:

- 24/7 operation (no sleep, no breaks)
- Massively parallel exploration
- Automatic verification loops
- Could 10x 100x research output

The question: This doesn't mean mathematicians are out of a job next week. It means the workflow changes.

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The Bottom Line

What Aletheia achieved:

- **5 Erdős problems** solved autonomously (open for decades)
- **1 complete research paper** (zero human intervention)
- **93.1% IMO-Proofbench** (previous SOTA: 65.7%)
- **6/10 FrontierMath** challenge problems solved
- **10% compute reduction** for same IMO accuracy
- **PhD level performance** on FutureMath Basic

What it means:

For mathematicians:

- New research assistant (tireless, thorough)
- Accelerated exploration of problem space
- Faster validation of conjectures
- Changed workflow, not replacement

For AI research:

- Proof of autonomous discovery capability
- Scaling laws extend beyond Olympiad level
- Generator Verifier Revisor pattern works
- Natural language proofs viable

For science:

- First AI generated peer reviewed math paper
- Research velocity could increase 10x 100x
- Human AI collaboration model validated
- Mathematical AGI boundary pushed significantly

The hard truth:

Competition math + Research math seemed like an impossible gap.

DeepMind has been building toward this since AlphaGo in 2016. Iron mathematical behavior into a schedulable search problem.

They did it. Aletheia is proof.

What's next:

Short term:

- More Erdős problems solved
- Broader mathematical domains
- Integration with formal proof systems

Long term:

- Autonomous research across all sciences
- AI generated knowledge acceleration
- Human AI research teams standard practice

The philosophical question:

When AI can discover original mathematics autonomously, have we created something that truly understands? Or just something that searches solutions space extremely well?

The narration points out what human mathematicians immediately notice: that initial "pick the right function" move feels like experience — the kind of simplifying direction people learn after years of wrestling with similar problems.

Maybe the answer doesn't matter. What matters: mathematical knowledge advances faster than ever before.

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Aletheia: The Greek goddess of truth. The AI that discovers it.

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Feng, T., et al. (2026). Towards Autonomous Mathematics Research. arXiv:2602.10177.

Feng, T., et al. (2026). Aletheia tackles FrontierMath autonomously. arXiv:2602.21201.

Google DeepMind (2026). Accelerating Mathematical and Scientific Discovery with Gemini Deep Think. Official Blog.

Code, prompts, and full retrospec: https://github.com/google-deepmind/superhuman/tree/main/aletheia

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What are your thoughts?

ashish Mar 25

This is fascinating. The fact that Aletheia can tackle problems that stumped mathematicians for decades suggests something fundamental about how AI can navigate mathematical structure in ways that supercomputers alone couldn't. Waiting for a peer?

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